Executive Summary

The objective of this portfolio project was to design and actively manage a portfolio capable of outperforming the S&P 500 over a 4 month trading period. As a team, we adopted a hybrid strategy that combined momentum investing with a passive management core, balancing upside potential of strong short-term trends with the stability of long-term diversification exposure through ETFs and bluechip companies. This hybrid approach was to capture market upside while managing downside risk through diversification. In our case, we wanted to use ETFs to diversify and manage our risks while exploiting short-term price for alpha generation.

Momentum Trading Strategy

We allocated 30% of our portfolio to index-based ETFs (VOO, QQQ, and GDX), holding these positions consistently as the passive, stabilizing portion of the portfolio. The remaining 70% was invested in individual equities that demonstrated strong momentum characteristics, including TSLA, NVDA, META, AAPL, UNH, WMT, RKT, AFRM, JPM, and BLK. This structure allowed us to pursue both stability and growth: ETFs anchored the portfolio during volatile periods, and momentum names offered the opportunity to outperform the market in the short run.

Significant Underperformance and Elevated Risk Characterize Portfolio's Results

Our portfolio ended with a total return of -1.92% (-5.98% annualized) which significantly underperformed the S&P500's annualized return of 26.23%. Our portfolio exhibited higher volatility with an annualized standard deviation of .207 compared to the S&P500’s .124, resulting in a portfolio beta of 1.2585. This indicates that our portfolio was 26% more volatile than the market which exposed us to greater systematic risk without the following returns.

Individual holdings performance was very scattered, ranging from our most impressive investment of GDX which had a return of 24.56% to our lowest loss in NVDA at -6.29%. Our three ETF positions all generated positive returns, which helped cushion losses from underperforming equity positions; however, these gains were insufficient to offset our losses in BLK (-3.28%), NVDA (-6.29%), UNH (-5.66%), and JPM (-0.29%). Within our equity investments, TSLA was our largest winner at a gain of 11.35%.

What Worked During Our Holding Period

Our ETF component of our strategy worked as planned, providing downside protection and positive returns from all three ETF holdings. GDX, our gold miners ETF, was our single best-performing investment at +24.56%, benefiting from falling interest rate expectations, a weaker dollar, and increased safe-haven demand. Our two other ETFs were VOO and QQQ, and while generating more modest returns of +0.54% and +0.02%, they provided stability and broad market exposure. These positions anchored our portfolio during volatile periods and demonstrated the value of passive indexing, even if their performance was marginal relative to the overall portfolio loss.

What Didn't Work and What we Would Do Differently Next Time

Our momentum strategy did not work due to two factors: poor security selection and suboptimal timing of both entries and exits. Our NVDA buy exemplifies this problem as we initially moved into profitable territory with the stock reaching highs of $212.19, but we failed to take profits and subsequently experienced a -6.29% loss when the stock declined post-earnings, despite strong fundamentals. Similarly, we sold RKT, HIMS, AFRM, UNH, WMT, and TEM at losses, demonstrating our inability to properly manage losing positions. The momentum investing philosophy requires rapid response to trend reversals, yet we allow small losses to become larger ones. We were not sufficiently active or attentive to current financial events and technical signals to effectively implement this strategy.

If we were to do this project again, one of the first changes would be to implement stricter and more clearly defined decision-making rules around our momentum strategy. Rather than relying on general sentiment, we could establish predetermined entry and exit criteria that are grounded in both technical indicators, analysis, and financial news. This would ensure that our trades align with a disciplined framework instead of being reactive. All in all, establishing a proactive stance would help us respond faster to trend reversals, reduce drawdowns, and enhance overall portfolio stability to capitalize on more upside.

Collaborative Structure Fostered Diverse Views but Hindered Timely Execution

Our team's organizational structure both enabled and constrained our portfolio management process. The collaborative approach ensured diverse perspectives and reduced individual biases in decision making. Different team members brought unique viewpoints, such as favored technology focused ETFs like QQQ while others preferred more diversification with VOO. The consensus driven approach also created significant challenges for momentum investing, which requires more rapid, decisive action. Time sensitive opportunities were harder to capitalize on when we had to coordinate between all our schedules to talk. Sometimes we had to go without a group consensus and rely on one or two members to make a trade, or we would wait for the whole group consensus which would lose out on timing.

Conclusion

Although our portfolio did not outperform the S&P 500, the project provided valuable insight into the realities of active management. The hybrid strategy was conceptually strong and aligned with FE445 principles: diversification, beta control, and momentum-based alpha opportunities. However, our performance reflected the operational challenges of momentum investing, timing, monitoring, and discipline, especially when coordinated across a team with limited availability. This experience emphasized the importance of strong communication, structured decision frameworks, and risk management tools. If repeated with more active oversight and refined processes, this strategy has the potential to outperform the market while maintaining controlled risk.